

Practice Set 4

MA628 - Financial Derivatives Pricing

February 9, 2026

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1. A stock price is currently at Rs 40. It is known that at the end of 1 month stock price will be either Rs 38 or Rs 42. The risk-free interest rate is 8% per annum compounded continuously. Find the delta hedge for 1 month European call with strike 39. **Answer:** 0.75
 2. A “covered call” is defined as long position in underlying asset and short position in call option. Suppose a stock price is currently at 40 Rs. It is known that at the end of 1 month stock price will be either 38 Rs or 42 Rs. The risk-free interest rate is 8% per annum compounded continuously. Find the price of the 1 month covered call with strike 39. **Answer:** 38.33
 3. Suppose XYZ company’s stock price is currently \$50. In the next six months it will either fall to \$30 or rise to \$80. What is the option delta of a call option with an exercise price of \$50? **Answer:** 0.6
 4. Suppose ABC’s stock price is currently \$50. In the next six months it will either fall to \$40 or rise to \$60. What is the current value of a six-month call option with an exercise price of \$50? The six-month risk-free interest rate is 2% (assume simple compounding). **Answer:** 5.39